KENGANSOCIAL · MIRAISTACK · READINESS REPORT

Ready for 1,000 users?

End-to-end test, capacity model for 1,000 users on each plan and on a realistic mix, the cost to run it and the profit, and the deployment checklist. Prepared 23 October 2026 from commit `3304938` plus local changes.

Not yet
Today one server renders one video and one clip project at a time. At 1,000 users the queues grow faster than they empty during busy hours.

R490 047/month
profit for a typical mix of 1,000 users (88.1% margin); R395 076 if everyone uses 100% of their plan.

~11 paying users
cover the fixed and base server costs (R10 313/month) at the mix below.

What it takes: move clip projects and video renders off the API into a pool of worker machines that grows and shrinks with demand (about 14 workers at the busiest hour for a typical mix, up to 76 at full use), keep files in Cloud Storage instead of on the server's disk, and transcribe long videos with a speech API or GPU. The API itself already handles 1,000 users with room to spare. The remaining security, payment and approval items are in the checklist.

1. What was tested end to end

TEST (THIS LAPTOP: I5-12500H, 16 THREADS)	RESULT	ON ONE 4-VCPU CLOUD WORKER
Clip project: 8-minute podcast → transcribe, find moments with AI, render 5 captioned shorts	225 s, 1303 CPU-seconds	≈ 7 min (model predicted 7 min)
3D product ad, 15 s vertical, voiceover + effects	53 s, 333 CPU-s	≈ 2 min
3D product ad, 25 s landscape	93 s, 642 CPU-s	≈ 3 min
Launch video, 20 s (first render in a process)	386 s, 1424 CPU-s	includes ~35 s render-bundle build and an AI provider failing over
YouTube link → video file (yt-dlp)	Pass	19 s test video downloaded
API load: dashboard requests, 50 users at once	248 requests/s, p95 441 ms, 0 errors	1,000 users need ~40 requests/s at peak
API load: 100 users at once	196 requests/s, p95 1203 ms, 0 errors	Slower but correct; add a second API only after the scheduler is split out
API load: light endpoint, 200 users at once	6,018 requests/s, p95 48 ms	Express itself is not the limit; Firestore round trips are
Automated tests (video ads, security, client lint)	Pass	Video ads 13 + routes 1; client lint clean

> **Not tested here** (needs the real servers): Payfast checkout and renewals, posting to live social accounts, a full restore from backup, and a load test with real separate users from many locations. These are in the checklist.

2. Can it house 1,000 users today?

What already scales

- **API:** one process served 248 dashboard requests per second; 1,000 users need about 40 (20% online at peak, one request every 5 s). About 5× headroom.
- **Database:** Firestore scales on its own; costs are per read and write (counted in the model).
- **Accounts and workspaces:** every request is scoped to a workspace; 1,000 workspaces are not a storage problem.
- **Pricing:** every paid plan stays above 60% margin at full use (this report confirms 67–85% with a real fleet).

What breaks at 1,000 users

- **One clip project at a time** for the whole app (`server/src/routes/clips.ts:72`). A typical mix sends 6.3× more clip work in the busiest hour than one queue can finish.
- **One video render at a time** (launch, tutorial, product ads, carousels: `chrome-queue.js`, `routes/brag.ts:20`). Peak load 1.8× one worker.
- **Video service fixed at 3 jobs** (`python-service/main.py:63`) and shares a folder on the same machine with the API.
- **Long Agency videos on CPU:** a 120-minute source with the medium speech model takes over 2 hours on one worker.

Other gaps found during the test

- **Files on the server's disk:** render scratch files, the video-ad cache (`server/.cache/video-ads`) and the upload registry live on the VM. Workers on other machines need Cloud Storage. At full use 1,000 Agency users could store 20 TB.
- **Rate limit per IP address** (fixed 24 October): now per signed-in user in Redis, with a higher per-IP ceiling.
- **Render bundle built on first use:** the first video after a restart waits ~35 s longer. Build it into the Docker image.
- **AI providers:** the OpenRouter key is empty on this machine and a Gemini request failed during the launch-video test before another provider took over. Production needs two working AI providers and an alert when one fails.
- **Scheduler inside the API:** posting, webhooks and workers run in the one API process, so a second API copy would double-post. Fine for 1,000 users, but it is a single point of failure: split the scheduler into its own process first.
- **Voice and music accounts:** ElevenLabs is on the free tier (10,000 characters, about 27 ads). A typical mix needs ~504k characters and ~1 400 Treblo tracks a month.

3. 1,000 users on each plan

Each row is a separate test: 1,000 customers all on that plan. **Typical** means they use 35% of every allowance; **full** means 100% (the worst case the prices are built for). Revenue is the monthly list price without VAT; costs include Payfast fees, AI, voice, music, transcription, storage, Firestore, the worker fleet sized for the busiest hour, the API server, hosting, backups and licences. Exchange rate R17.5/US$, with VAT on imported services and a 20% buffer on provider prices.

Typical use (35% of allowances)

PLAN (1,000 USERS)	REVENUE EX VAT	PAYFAST FEES	PROVIDERS AI, VOICE, MUSIC, STORAGE	SERVERS + FIXED	PROFIT / MONTH	MARGIN	PEAK WORKERS RENDERS + CLIPS	TIME TO RESULT RENDER / CLIP PROJECT	STORAGE
Free free	R0	R0	R1 644	R7 863	R-9 506	–	2 + 0	2 min / –	0.2 TB
Creator Starter R299/month	R260 000	R13 303	R18 311	R10 313	R218 073	83.9%	3 + 3 STT: 3 + 2	2 min / 6 min STT: 5 min	0.7 TB
Creator Pro R699/month	R607 826	R28 023	R39 132	R10 407	R530 264	87.2%	5 + 9 STT: 5 + 4	2 min / 16 min STT: 5 min	1.7 TB
Agency R3 499/month	R3 042 609	R131 063	R195 806	R44 857	R2 670 883	87.8%	14 + 61 STT: 14 + 8	2 min / 68 min STT: 7 min	6.8 TB
Enterprise R8 999/month	R7 825 217	R333 463	R195 806	R44 857	R7 251 091	92.7%	14 + 61 STT: 14 + 8	2 min / 68 min STT: 7 min	6.8 TB

Full use (100% of allowances)

PLAN (1,000 USERS)	REVENUE EX VAT	PAYFAST FEES	PROVIDERS AI, VOICE, MUSIC, STORAGE	SERVERS + FIXED	PROFIT / MONTH	MARGIN	PEAK WORKERS RENDERS + CLIPS	TIME TO RESULT RENDER / CLIP PROJECT	STORAGE
Free free	R0	R0	R3 982	R7 863	R-11 844	–	7 + 0	8 min / –	0.5 TB
Creator Starter R299/month	R260 000	R13 303	R53 689	R18 726	R174 282	67.0%	17 + 14 STT: 17 + 12	7 min / 22 min STT: 19 min	2.0 TB
Creator Pro R699/month	R607 826	R28 023	R115 720	R48 391	R415 691	68.4%	34 + 49 STT: 34 + 26	7 min / 41 min STT: 21 min	4.9 TB
Agency R3 499/month	R3 042 609	R131 063	R602 152	R276 311	R2 033 082	66.8%	123 + 343 STT: 123 + 59	7 min / 2.4 h STT: 24 min	19.5 TB
Enterprise R8 999/month	R7 825 217	R333 463	R602 152	R276 311	R6 613 291	84.5%	123 + 343 STT: 123 + 59	7 min / 2.4 h STT: 24 min	19.5 TB

> **Reading the table:** "STT" shows the same plan with cloud transcription (speech API) for clip projects. "Peak workers" is the number of 4-vCPU worker machines needed in the busiest hour so that 95% of renders start within 60 s and 90% of clip projects start within 5 min. Workers shut down when idle; the cost column pays for busy time (60% average utilisation) with at least one worker per queue always on. Free users cost R10–R12 each per month including the base servers. Enterprise is priced from the Agency allowances.

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4. A realistic mix of 1,000 users

Assumed mix: 500 Free, 250 Creator Starter, 150 Creator Pro, 80 Agency, 20 Enterprise.

MIXED 1,000 USERS	REVENUE EX VAT	FEES + PROVIDERS	SERVERS	FIXED	PROFIT / MONTH	MARGIN	PEAK WORKERS	TIME TO RESULT
Typical use (35%)	R556 087	R55 534	R7 894	R2 613	R490 047	88.1%	4 + 10	2 min / 37 min
Typical, spot workers	R556 087	R55 534	R4 583	R2 613	R493 358	88.7%	4 + 10	2 min / 37 min
Typical, cloud transcription	R556 087	R62 266	R7 700	R2 613	R483 508	86.9%	4 + 3	2 min / 6 min
Full use (100%)	R556 087	R117 670	R40 728	R2 613	R395 076	71.0%	27 + 49	7 min / 84 min
Full, spot workers	R556 087	R117 670	R16 075	R2 613	R419 729	75.5%	27 + 49	7 min / 84 min
Full, cloud transcription	R556 087	R156 141	R21 686	R2 613	R375 648	67.6%	27 + 14	7 min / 22 min

Monthly cost, typical use

Payfast fees	R24 684
AI credits	R10 143
Clip AI (transcript ranking)	R3 013
Voice, music, render licence	R5 342
Post delivery (egress)	R6 466
Storage (GB-month + download)	R3 756
Firestore and sync	R2 130
Worker VMs (autoscaled)	R5 094
API VM (with Redis, Caddy)	R2 450
Web hosting	R350
Remotion licence minimum ($100, render fees count towards it)	R2 012
Backups, snapshots, domain, email, monitoring	R600
Total cost	**R66 040**
Revenue (ex VAT)	R556 087
Profit	**R490 047**

Monthly cost, full use

Payfast fees	R24 684
AI credits	R28 980
Clip AI (transcript ranking)	R15 988
Voice, music, render licence	R15 263
Post delivery (egress)	R18 475
Storage (GB-month + download)	R10 731
Firestore and sync	R3 550
Worker VMs (autoscaled)	R37 928
API VM (with Redis, Caddy)	R2 450
Web hosting	R350
Remotion licence minimum ($100, render fees count towards it)	R2 012
Backups, snapshots, domain, email, monitoring	R600
Total cost	**R161 011**
Revenue (ex VAT)	R556 087
Profit	**R395 076**

> **Cloud transcription** (a speech API at US$0.006/min instead of Whisper on the worker's CPU) costs slightly more but cuts clip workers at peak from 10 to 3 (typical) and Agency clip projects from 68 min to 7 min. Recommended for Creator Pro and Agency. **Spot workers** (interruptible VMs, ~35% of the price; check availability in Johannesburg) save R24 653 a month at full use.

5. Target setup for 1,000 users

PART	TODAY	FOR 1,000 USERS
API	One container on one VM, also runs the scheduler, webhooks and renders	One API VM (e2-standard-4) serving requests only; scheduler in its own single process
Clip projects	One at a time, in the API	Durable job queue (BullMQ on Redis, already used for posts and AI edits) and a clip worker pool, 10–49 at peak, scaling to zero spare
Video renders	One at a time per API process	Render worker pool, 4–27 at peak; render bundle built into the image
Transcription	Whisper on CPU in the video service	Speech API (or a GPU worker) for sources over 20 minutes; CPU Whisper for short ones
Files	Shared folder on the VM's disk	Cloud Storage (Firebase bucket) for uploads, sources and outputs; workers keep only scratch files; lifecycle rules delete expired renders
Rate limits	Per IP, in memory	Per user, in Redis
Monitoring	Container logs	Uptime check on `/api/health/ready`, queue length and wait-time alerts, error tracking

6. Deployment checklist

MUST blocks launch to paying users. SCALE is needed before ~100 active users (single queues get slow). SOON can follow in the first weeks.

A. Code changes for 1,000 users

- [ ] SCALE Run clip projects through a durable queue (BullMQ) with separate worker processes; remove the one-at-a-time queue in `routes/clips.ts`.
  Acceptance: 20 clip projects submitted at once finish in parallel across workers; a restarted worker resumes or fails the job cleanly with a refund.
- [ ] SCALE Move launch/tutorial videos, product ads and carousels to render workers; keep Chrome off the API process.
  Acceptance: the API stays under 500 ms p95 while 10 renders run.
- [ ] SCALE ~~Make the video service concurrency configurable~~ Done 24 Oct: `MAX_CONCURRENT_JOBS` (set 1 on a dedicated worker VM).
- [ ] SCALE Store uploads, clip sources and outputs in Cloud Storage; workers download to scratch disk and upload results. Move the video-ad cache and upload registry off the VM disk.
- [ ] SCALE Transcribe long sources with a speech API or GPU worker; keep CPU Whisper for short videos.
  Agency 120-minute sources drop from ~2.4 h to ~25 min end to end.
- [ ] SCALE Split posting, webhooks and token refresh into one scheduler process, so the API can run more than one copy.
- [ ] SCALE ~~Rate-limit per signed-in user in Redis~~ Done 24 Oct: per user (1,000 per 15 min) plus a per-IP ceiling (10,000), shared through Redis.

- [ ] SOON Build the Remotion bundle and the Whisper model into the Docker images.
- [ ] SOON Render the 10 clips of a project in parallel across workers (full-use clip projects are mostly render time).

B. Security and billing (from the 23 October evaluation, still open)

- [ ] MUST Private media: stop serving uploads anonymously by URL; use signed, expiring links.
  Listed as the remaining security blocker in DEPLOYMENT_FIX_PROGRESS.
- [ ] MUST Payfast sandbox on the live server: subscribe, renew, failed payment, cancel; then one real payment and refund.
- [ ] MUST Replace the Firebase service-account key that was pasted in chat; rotate the ElevenLabs and Treblo keys pasted in chat today.
- [ ] MUST Production settings: `NODE_ENV=production`, no `DEV_PLAN_OVERRIDE`, real `SERVER_URL` / `CLIENT_URL`, `TOKEN_ENCRYPTION_KEY`, `FREE_CLIP_SECRET`, Meta webhook secrets.
- [ ] SOON Run the containers as a non-root user; set memory limits per container.

C. Servers and accounts

- [ ] MUST Re-open Google Cloud billing for `kengansocialv2`; create the API VM and point `api.kengansocial.co.za` at it.
- [ ] MUST Web app hosting for `kengansocial.co.za` (new Vercel project or served from the VM).
- [ ] MUST Deploy Firestore rules and indexes and Storage rules; Firestore on the pay-as-you-go plan.
- [ ] SCALE Worker instance template + autoscaling group (e2-standard-4, 150 GB disk) for clip and render queues; start with 14 max and raise with demand. Check spot VM availability in africa-south1.
- [ ] MUST AI: two working providers (OpenRouter and Gemini) with billing; budget ~R10 143/month for a typical mix.
- [ ] MUST ElevenLabs paid plan sized for ~504k–1 440k characters/month; Treblo credits for ~1 400–4 000 tracks/month.
- [ ] MUST Remotion company licence (rendering is a commercial use).
- [ ] MUST Mailboxes: billing@, support@, privacy@kengansocial.co.za receive mail.

D. Platforms and legal

- [ ] MUST Meta app review (Advanced Access, business verification); webhooks subscribed; Business Suite auto-replies off.
- [ ] MUST TikTok and YouTube app approval. The YouTube-link import is on: keep the ownership confirmation, and consider switching it off during the reviews.
- [ ] MUST POPIA: register the Information Officer; review the Privacy Policy and Terms (Payfast, renewal) with someone qualified.

E. Prove it on staging before inviting users

- [ ] MUST Run `node scripts/check-deploy.mjs` against the real domains; sign up two workspaces and test cross-workspace denial.
- [ ] MUST Load test on staging: 1,000 simulated users (k6 or autocannon from several machines), plus 50 clip projects and 50 renders submitted in one hour. Target: dashboard p95 under 1 s, render wait p95 under 60 s, clip wait p90 under 5 min.
- [ ] MUST Backups: scheduled Firestore exports and disk snapshots; restore into a separate project and time it.
- [ ] MUST Monitoring: uptime on `/api/health/ready`, alerts on queue wait, failed jobs, AI provider errors and disk space.
- [ ] SOON Measure job times on the real workers and update `scripts/capacity-model.mjs` and `scripts/pricing-model.mjs`; rebuild this report.

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7. How the numbers were made

- Job times were measured on this laptop on 23 October 2026 and converted to a 4-vCPU worker: the larger of the measured time and CPU-seconds × 1.2 ÷ 4. The clip-project formula in the pricing model matched the measurement within 10%.
- Typical use = 35% of each allowance; full use = 100%. Typical clip sources are half the plan's maximum length; full use takes the maximum, 10 clips of 60 s.
- Busiest hour = 2× the average working hour (22 days × 10 hours). Worker counts use a standard queueing formula (Erlang C) for the wait targets above.
- Prices: monthly list prices including VAT; Payfast 3.2% + R2 plus VAT on the fee; provider prices from `scripts/pricing-model.mjs` with VAT on imported services and a 20% buffer; e2-standard-4 at US$140/month; spot at 35% of that (estimate).
- Annual billing (10% off) lowers revenue by 10%; every paid plan stays above 60% margin on annual prices at full use (pricing model).
- Re-run: `node scripts/capacity-model.mjs` and `node scripts/build-capacity-doc.mjs`.

KenganSocial is operated by MiraiStack (registration K2025971590). Figures are estimates for planning; confirm provider prices and measure on the real servers before relying on them.